

ASYN4 Machine Performance Report

Field	Value
NAME	RS MOTOR
REKE	SC315L
DATUM	24.09.26

Partial-Load Performance

Load	P_mech (kW)	Eta	cos(phi)	I1 (A)	Slip (%)	Pcu1 (kW)	Pcu2 (kW)	M (Nm)
1/4	22.50	0.8808	0.415	49.20		0.466	0.161	360.0
2/4	45.16	0.9230	0.636	61.59		0.730	0.567	726.9
3/4	67.39	0.9311	0.741	78.14		1.175	1.262	1091.2
4/4 (rated)	89.90	0.9293	0.790	97.92		1.846	2.316	1465.3
5/4	112.75	0.9223	0.808	120.81		2.810	3.842	1851.8

Breakdown / Locked-Rotor Summary

Quantity	Value
Breakdown slip (Kipp)	10.793 %
Breakdown torque	3247.3 Nm
Locked-rotor torque (theoretical)	2784.5 Nm
Locked-rotor current (theoretical)	427.5 A
Locked-rotor torque (corrected, KKA=0.50)	3396.6 Nm
Locked-rotor current (corrected)	471.8 A
Time constant TK	0.0384 s
Time constant THAAPT (rotor)	0.2565 s
Time constant TSTREU (leakage)	0.0069 s

Cooling & Material Information

Quantity	Value
Dominant cooling path (IKLINK)	0
Stator yoke iron area AR1	0.03944 m2
Stator tooth iron area AZ1	0.02373 m2
Rotor yoke iron area AR2	0.07443 m2
Rotor tooth iron area AZ2	0.02660 m2